

Roll no.

MID SEMESTER EXAMINATION MAR 2025

Name of the Program: B. Tech

Semester: II

Name of the Course: Basic Electronics Engineering

Course Code: TEC 201

Time: 90mins

Maximum Marks: 50

Note:

- (i) Answer **all the questions** by choosing **any one of the sub questions**.
- (ii) Each question carries 10 marks

Q1		(10 marks)	CO1
a)	Simplify the following Boolean expression and realize the simplified expression with basic logic gates. (i) $Y = AB + AC + B(B + C)$ (ii) $Y = A'B'C + A'BC' + A'B + B'C'$		
OR			
b)	Convert the following bases: (i) $(3FF)_H = (?)_{10} = (?)_2$ (ii) $(3754)_8 = (?)_{10} = (?)_2 = (?)_{16}$		
Q2		(10 marks)	CO1
a)	Determine the base of the following numbers: (i) $(210)_3 = (25)_7$ (ii) $(1DD)_{16} = (735)_7$		
OR			
b)	Implement XOR and XNOR using NAND and NOR gate only.		
Q3		(10 marks)	CO1
a)	Solve the following expression using K-Map and implement using NAND gate only: $F(A,B,C,D) = \sum m(4, 5, 6, 7, 15) + \sum d(11, 13)$		
OR			
b)	Solve the following expression using K-Map and implement using NOR gate only: $F(A,B,C,D) = \sum m(0, 1, 2, 5, 13, 15)$		
Q4		(10 marks)	CO2
a)	Differentiate Conductor, Semiconductor and Insulator based on energy band diagram. Explain Intrinsic and Extrinsic semiconductors as well.		
OR			
b)	The intrinsic carrier concentration for silicon at room temperature is 1.5×10^{10} per cm^3 . If the mobility of electrons and holes are $1300 \text{ cm}^2/\text{V-s}$ and $450 \text{ cm}^2/\text{V-s}$ respectively. What is the conductivity of intrinsic silicon? If silicon is doped with 10^{18} boron atoms per cm^3 , what is its conductivity?		
Q5		(10 marks)	CO2
a)	Determine the conductivity of intrinsic and extrinsic semiconductor.		
OR			
b)	A specimen of germanium at 300°K for which the density of carriers is 2.5×10^{13} per cm^3 , is doped with impurity atoms such that there is one impurity atom for 10^6 germanium atoms. All the impurity atoms may be assumed ionized. The conductivity of doped material is 25.64 S/cm . Carrier mobility for germanium at 300°K is $3600 \text{ cm}^2/\text{V-s}$. For the doped material, find the electron and hole densities. $e = 1.602 \times 10^{-19} \text{ C}$.		

